

FIFA World Cup Dataset

Data Dictionary

This document describes every table and field in the FIFA World Cup five-year analysis dataset (2002–2018), as modeled in the Power BI star schema: one fact table (Facts_Results) and two dimension table (Teams, Years).

1. Data Model Overview

The dataset follows a star-schema design. Facts_Results holds one row per country per tournament (160 rows total). Teams holds one row per unique country (59 rows). Years holds one row per tournament (5 rows). Both dimension tables have a one-to-many relationship into Facts_Results.

Table	Type	Row Count	Grain
Facts_Results	Fact	160	One row per team per tournament
Teams	Dimension	59	One row per unique country
Years	Dimension	5	One row per tournament (2002–2018)

2. Table: Facts_Results

The central fact table. Each row represents one country's complete record for a single tournament.

Field Name	Data Type	Description
Year	Whole Number	The tournament year. One of: 2002, 2006, 2010, 2014, 2018. Foreign key to Years[Year].
Position	Whole Number	The country's final ranking position within that tournament's results table (1 = highest-ranked).
Team	Text	The name of the competing country. Foreign key to Teams[Team].
Games Played	Whole Number	Total matches played by the team in that tournament.
Win	Whole Number	Number of matches won in that tournament.
Draw	Whole Number	Number of matches drawn in that tournament.
Loss	Whole Number	Number of matches lost in that tournament.

Field Name	Data Type	Description
Goals For	Whole Number	Total goals scored by the team in that tournament.
Goals Against	Whole Number	Total goals conceded by the team in that tournament.
Goal Difference	Whole Number	Goals For minus Goals Against. Can be negative. Source data used a non-standard Unicode minus character, corrected during cleaning — see Section 5.
Points	Whole Number	Tournament points (3 per win, 1 per draw, 0 per loss, per standard group-stage scoring).

3. Table: Teams

A distinct, de-duplicated list of every country that appeared in at least one of the five tournaments studied.

Field Name	Data Type	Description
Team	Text	Unique country name. Primary key. 59 distinct values, derived from Facts_Results[Team] with duplicates removed.

4. Table: Years

A distinct list of the five tournament years covered by this analysis.

Field Name	Data Type	Description
Year	Whole Number	Tournament year. Primary key. Five distinct values: 2002, 2006, 2010, 2014, 2018, derived from Facts_Results[Year] with duplicates removed.

5. Data Quality Notes

Goal Difference Unicode minus character: The original source files represented negative goal differences using a non-keyboard Unicode minus sign rather than a standard hyphen. This caused the column to import as Text rather than a numeric type. It was corrected using Power Query's Replace Values (or an equivalent Text.Replace + Number.FromText custom column) before the column was converted to Whole Number.

Nigeria missing 2006 row: Nigeria did not qualify for the 2006 tournament and therefore has no row in Facts_Results for Year 2006. This is expected and correct; it was not treated as missing data requiring imputation.

Team name consistency: Team names are matched exactly (case and spelling-sensitive) between Facts_Results and Teams. Historical naming variants (e.g., "Serbia and Montenegro" appearing only in earlier tournaments, distinct from "Serbia" in later ones) are treated as separate countries, consistent with how they competed at the time.

6. Key Measures Built on This Model

Measure	Definition (plain language)
Total Games Played	Sum of Games Played across the current filter context.
Total Wins / Losses / Draws	Sum of Win / Loss / Draw across the current filter context.
Total Goals For / Against	Sum of Goals For / Goals Against across the current filter context.
Total Points	Sum of Points across the current filter context.
Countries Played	Distinct count of Teams[Team] represented in the current filter context.
Appearances	Distinct count of Facts_Results[Year] for a given team how many of the 5 tournaments they played in.
Win Rate	Total Wins ÷ Total Games Played.
Avg Goals Per Game	Total Goals For ÷ Total Games Played.
Highest / Lowest Country (per metric)	The team returning the maximum/minimum value of a given measure, with an alphabetical tiebreaker where values are equal.

7. Table: Metrics

A disconnected helper table used primarily for dynamic report visuals and KPI selection.

Field	Data Type	Description
Metric Name	Text	Stores the names of selectable performance metrics used in visuals.

Measures

Measure	Description
Highest Country	Returns the country with the highest value for the selected metric.
Highest Values	Returns the corresponding highest metric value.
Lowest Country	Returns the country with the lowest value for the selected metric.
Lowest Values	Returns the corresponding lowest metric value.

8. Relationships

The model follows a **star schema**.

From	To	Cardinality	Cross-filter
Years[Year]	Facts_Results[Year]	One-to-Many	Single Direction
Teams[Countries]	Facts_Results[Team]	One-to-Many	Single Direction

The **Metrics**, **Performance Measures**, and **Total Measures** tables are measure/helper tables and do **not** participate in model relationships.

9. Model Design Notes

The semantic model has been organized into:

- **1 Fact table**
 - Facts_Results
- **3 Dimension/Helper tables**
 - Years
 - Teams
 - Metrics
- **2 Measure tables**
 - Total Measures
 - Performance Measures

Separating measures into dedicated measure tables improves report organization, makes the Fields pane easier to navigate, and follows Power BI modeling best practices.

10. Key Performance Indicators (KPIs)

The report provides KPIs including:

- Total Countries
- Total Games Played
- Total Wins
- Total Draws
- Total Losses
- Total Goals Scored
- Total Goals Conceded
- Total Goal Difference
- Total Points
- Tournament Appearances
- Win Rate (%)
- Average Goals per Game
- Average Points per Game
- Highest Goals
- Highest Wins
- Highest Points
- Best Goal Difference
- Highest Appearance
- Lowest Appearance
- Highest Country by Selected Metric
- Lowest Country by Selected Metric

